

PHOENIX SKY HARBOR, AZ, USA

WMO: 722780

Lat: 33.428N

Lon: 112.004W

Elev: 1107

StdP: 14.12

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 23183

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	39.3	41.9	2.4	6.5	66.3	6.9	8.2	66.6	19.7	56.3	16.9	59.0	3.5	100	0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	110.5	69.2	108.5	69.0	106.7	68.9	75.5	93.6	74.8	93.8	74.0	94.1	9.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.5	121.1	81.9	69.5	113.2	84.1	67.6	105.8	86.4	39.7	92.8	38.9	93.6	38.2	93.6	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.2	16.7	14.2	DB	34.5	114.9	2.7	2.4	32.5	116.6	30.9	117.9	29.4	119.3	27.4	121.0
			WB	28.6	77.4	2.6	1.9	26.7	78.7	25.1	79.8	23.7	80.8	21.7	82.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.5	56.9	59.7	66.4	72.7	81.5	91.4	95.3	94.0	88.9	76.9	65.2	55.9	(6)
	DBStd	15.27	5.44	5.98	6.60	6.95	6.55	5.38	4.41	4.37	5.06	6.76	6.67	5.38	(7)
	HDD50	27	10	4	0	0	0	0	0	0	0	0	1	12	(8)
	HDD65	874	255	165	64	19	1	0	0	0	0	6	79	285	(9)
	CDD50	9328	223	275	510	682	977	1243	1403	1365	1167	833	456	194	(10)
	CDD65	4698	3	16	108	250	513	793	938	900	717	374	84	2	(11)
	CDH74	72041	52	214	1239	2990	7255	13332	16027	15047	10868	4222	769	26	(12)
	CDH80	46794	1	29	407	1351	4234	9286	11605	10637	6948	2082	213	1	(13)
(14) Wind	WSAvg	6.2	4.8	5.5	6.2	7.3	7.2	7.1	7.5	6.8	6.3	5.6	4.9	4.6	(14)
(15) Precipitation	PrecAvg	7.4	0.8	0.7	0.9	0.2	0.1	0.1	0.9	0.9	0.7	0.6	0.6	0.9	(15)
	PrecMax	15.2	5.2	3.1	3.1	1.3	1.2	1.7	5.2	3.5	3.3	4.4	2.8	4.0	(16)
	PrecMin	2.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	2.9	0.9	0.8	0.8	0.3	0.3	0.2	0.9	0.9	0.8	0.8	0.7	0.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.8	83.8	92.6	97.9	106.4	113.6	113.6	112.1	108.0	100.1	89.6	77.0	(19)
		MCWB	52.9	54.7	58.7	60.5	64.9	68.1	70.0	70.4	69.2	65.3	58.5	53.3	(20)
	2%	DB	75.1	79.9	87.7	94.4	102.6	109.5	110.8	109.5	105.4	96.6	85.5	73.7	(21)
		MCWB	51.6	53.9	57.1	59.2	63.2	66.1	70.0	70.8	68.6	63.9	57.6	52.6	(22)
	5%	DB	71.9	76.4	84.6	91.2	99.5	107.5	108.7	107.3	103.2	93.8	82.1	70.4	(23)
		MCWB	50.7	52.9	56.0	58.0	61.9	65.8	70.1	70.8	68.2	62.7	56.7	51.2	(24)
	10%	DB	68.7	72.9	81.3	87.9	96.6	105.3	106.5	105.1	100.7	90.8	78.5	67.2	(25)
		MCWB	49.9	52.0	54.8	56.9	61.0	65.2	70.0	71.0	68.3	61.6	55.5	49.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.4	59.1	60.5	62.9	67.2	73.1	76.9	77.2	75.8	71.8	62.9	58.9	(27)
		MCDB	65.6	71.7	83.3	91.7	97.2	99.9	93.4	96.3	91.3	85.1	77.7	67.4	(28)
	2%	WB	55.7	57.3	59.1	60.9	65.6	71.3	75.4	75.7	74.3	68.2	60.8	56.2	(29)
		MCDB	65.2	69.6	81.4	89.6	95.6	98.7	93.4	94.5	91.5	86.9	75.7	66.2	(30)
	5%	WB	53.8	55.7	57.5	59.3	64.0	69.8	74.7	75.0	73.2	66.2	59.0	54.2	(31)
		MCDB	65.0	69.2	78.7	86.4	94.3	98.1	93.8	94.4	91.7	86.3	75.6	64.5	(32)
	10%	WB	52.1	54.0	56.1	57.9	62.3	68.0	73.8	74.2	72.1	64.2	57.2	52.4	(33)
		MCDB	64.3	68.4	76.8	84.7	92.7	98.9	94.1	94.6	91.3	85.3	75.1	63.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	20.7	20.8	22.8	23.2	23.7	24.3	20.6	20.0	20.8	22.3	21.5	19.8	(35)
		MCDBR	25.3	26.4	27.5	27.4	26.8	26.0	22.2	21.8	22.9	25.2	25.7	24.0	(36)
	5% WB	MCWBR	10.7	10.8	9.6	9.6	8.6	7.6	6.0	5.6	6.6	8.9	10.0	10.2	(37)
		MCWBR	18.6	21.0	24.6	25.1	24.7	21.6	19.5	19.5	18.3	19.8	21.2	18.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.284	0.290	0.317	0.326	0.335	0.346	0.408	0.394	0.350	0.325	0.294	0.284	(40)
		taud	2.555	2.547	2.450	2.413	2.419	2.388	2.346	2.373	2.467	2.487	2.531	2.559	(41)
	Ebn at Noon		299	308	305	305	301	296	277	280	290	290	292	293	(42)
		Edh at Noon	25	29	34	37	37	38	40	38	33	30	26	24	(43)
(44) All-Sky Solar Radiation	RadAvg		1075	1378	1853	2305	2585	2678	2348	2163	1938	1561	1192	974	(44)
	RadStd		80	107	88	63	86	98	103	69	72	89	51	51	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.95	N/A	N/A	+0.47	N/A	N/A	N/A	-106	+362	+252	
(47) Regional (13 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	N/A	-146	+176	+54	

Nomenclature: See separate page